



A fusion of technology and fashion

Infineon Technologies AG has collaborated with Adidas AG to introduce the Lighting Shoe. The Adidas NMD S1 shoe, equipped with advanced semiconductor technology, detects



Infineon Lighting Shoe in collaboration with Adidas (Credit: <https://www.infineon.com>)

environmental music and rhythms to create customisable LED lighting displays. It uses XENSIV MEMS microphones to capture audio signals, processed by a PSoC microcontroller, and transformed into vibrant LED illuminations. This integration showcases the fusion of technology and fashion, where the Infineon Lighting Shoe prototype highlights the potential of semiconductor technology. The shoe also incorporates EZ-PD BCR technology for USB-C Power Delivery charging and is powered by efficient OptiMOS power MOSFETs. All these technological elements are seamlessly integrated into the original Adidas NMD S1 design, ensuring that the shoe maintains its style and comfort.

World's first customisable e-textiles with graphene multimodal technology

Researchers at Korea Institute of Machinery and Materials (KIMM) and Korea Advanced Institute of Science and Engineering (KAIST) have developed the world's first graphene-based customisable electronic textiles (e-textiles). Using laser direct patterning technology, this eco-friendly technique avoids harmful chemicals and optical masks, allowing for precise patterns. The resulting e-textiles, featuring laser-induced graphene (LIG), have high electrical conductivity and potential applications in industrial and military clothing for health monitoring and custom smart garments in healthcare.

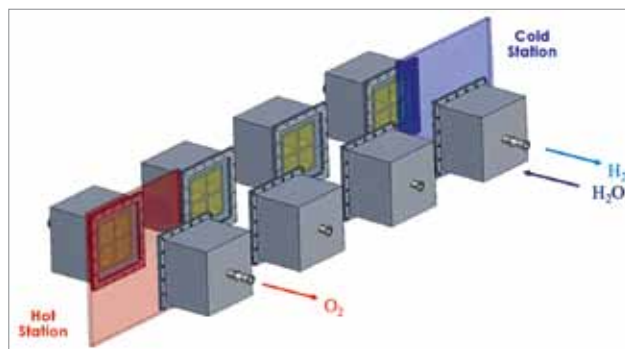
Adapting semiconductor techniques for enhanced battery development

Researchers at the US Department of Energy's Argonne National Laboratory have applied semiconductor industry methods to enhance solid-state batteries' charge capacity and streamline their manufacturing. By tailoring the chemistry of argyrodite surfaces using atomic layer deposition

from semiconductor manufacturing, they created a protective thin film on surfaces before pellet formation. This method uniformly coated electrolyte powder with alumina, confirmed by X-ray absorption spectroscopy, without altering structure. Coated electrolytes in battery cells showed improved performance, reducing reactivity with lithium anodes, preventing dendrite formation, and doubling ionic conductivity, potentially extending battery lifespan.

Solar powered system targets 40% solar heat utilisation for hydrogen production

MIT engineers have developed a train-like reactor system that revolutionises hydrogen production, utilising solar energy to capture 40% of solar thermal energy. This breakthrough enhances output, reduces costs, and contributes to



Unlike conventional hydrogen production systems reliant on fossil fuels, this new system exclusively harnesses solar energy (Credit: <https://news.mit.edu>)

decarbonisation in transportation. The Solar Thermochemical Hydrogen (STCH) system integrates with existing solar heat sources, employing a circular arrangement of mirrors to focus sunlight on a central tower. The core of the STCH system involves a two-step thermochemical process for water splitting, optimising efficiency and eliminating the need for energy-intensive pumps. This method increases efficiency from 7% to 40%, potentially transforming continuous hydrogen production and sunlight-to-liquid fuel conversion.

Flexible supercapacitors for efficient energy storage

The Materials for Energy Storage and Optoelectronic Devices Group at Sanatana Dharma College, Alappuzha, India, has engineered a hybrid electrode-based flexible symmetric supercapacitor. This innovative hybrid electrode, created without binders, combines high-molecular-weight polyaniline (PANi) and vanadium pentoxide nanostructures, exhibiting outstanding electrochemical properties, cycle stability, and high energy density. This approach results in a flexible,